FISCO BCOS evidence

Generated by `pnpm fisco` (`web/scripts/fisco.ts`) at 2026-10-20T12:42:38.032Z from `web/evidence/fisco/run-2026-10-20T11-21-48-467Z.json`. Do not edit by hand.

The same engine that commits to Base Sepolia ran end to end on a FISCO BCOS chain, WeBank's consortium blockchain, through its Web3 JSON-RPC. Every row was fetched back from the node while this file was written: status, block and gas come from the receipt. For every IntentLog commit the calldata was decoded and the contract read back, and the hash on FISCO equals the engine's intent hash. FISCO fails a call to an address without code, so where Base carries a money event as calldata on a self transfer, FISCO commits the event's digest to the IntentLog under the job id keccak("anchor"): one real contract call per event, a hash rather than the JSON. Amounts are demo units with no monetary value.

1. Chain

Field	Value
Node	FISCO BCOS 3.16.3, one node in Docker, group0, PBFT
Web3 chain id	20200
RPC	http://127.0.0.1:8545 (through `contracts/script/fisco_rpc_proxy.py`)
Compiler	solc 0.8.26, evm_version paris (no PUSH0), foundry profile `fisco`
Deployer	`0xC9E4C312E90cda14354340DFd7b9569d820C1201` (throwaway key for this chain)

2. Contracts

Contract	Address	Deploy transaction	Block	Gas
IntentLog	`0x9555c12c2789e44868070d0d877f9baadabe4f54`	`0xb1152d67…dd30`	71	13,783
Bond	`0xe3f2011697301e15e4cf829b0aeb189fab754029`	`0x2465a40f…ff72`	72	181,346
Lease	`0x9c46a913abaf37cacd67f01f6f3b3e70f7b66601`	`0xfbc07711…e50b`	73	74,181
JobBoard	`0x20dc56e80b73f6066208a9d8197748f861579d37`	`0xd97cdec1…92f2`	74	142,813

Smoke flow from `contracts/script/fisco.sh`, receipts fetched by the script. Native balances are switched on with the chain admin key `build_chain.sh` generates, so value moves for real: the vendor bond holds 5 units in Bond, the job fee of 60 units is locked in JobBoard and refunded on cancel, and the script checks every balance.

Call	Transaction	Block	Status
IntentLog.commit	`0xba66470cb8f03e9a84a6d28edda849c9355c44ae9748c06084e1552057268892`	76	success
Bond.postVendorBond	`0xd6a16deee7c1342224f07b768475623755bd382fe441322b19fada8c6cb7b8ac`	77	success
Bond.register	`0xddfdc559bdf4bb0e7f43013ce23c395afde6db140e9f2f0dc153efc3027a62f5`	78	success
Lease.open	`0x90e6634b35d395681b9a4c10ffc6cd25bb38b9db22f4d8ea4765aecaaf955cf6`	79	success
JobBoard.post	`0xf2abfe207dd864f860615fa2ae562f13ec9def4837143ee5c267af277fe1abba`	80	success
JobBoard.cancel	`0x1378d34e62c52f8f3e21a0e0555a09a29171859ff5c022dab4d98124c42af128`	81	success

3. A real run on FISCO: the fleet scenario (the five moments) and the neighbour scenario

FISCO is the only chain in this run. IntentLog used: `0xF86cED0C9738894079cA4FE9b48339b589d91990` (an earlier deployment on the same chain, before `fisco.sh` last redeployed; it is still live and was read back now).

78 transactions, 78 with a successful receipt in the recorded block, 30 IntentLog commits, 30 of them read back from the FISCO IntentLog with the engine's hash. 15 of these hashes also appear in the captured Base Sepolia run, 15 confirmed in the Base IntentLog by a read call: both ledgers hold the same EIP-712 intent hash under the same `commit(bytes32 jobId, bytes32 intentHash)` format.

#	Scenario	Event	Kind	Transaction	Block	Gas	Status	IntentLog check
1	fleet	anchor: vendor bond of 5000 yuan posted	vendor_bond	`0x0fddbcee…3a4e`	20	38,104	success	
2	fleet	anchor: fee 60 yuan locked for Spray 20 mu of maize, field north	lock	`0x739964fe…78b9`	21	38,104	success	
3	fleet	IntentLog.commit: int-001	log	`0xac221b10…9553`	21	38,104	success	hash matches, also on Base IntentLog
4	fleet	anchor: job cancelled, fee 60 yuan back to the farmer	refund	`0xe9ca272f…244d`	21	38,104	success	

#	Scenario	Event	Kind	Transaction	Block	Gas	Status	IntentLog check
5	fleet	anchor: fee 1500 yuan locked for Spray 100 mu of passion fruit, field north	lock	`0x939d5966…8380`	22	38,104	success	
6	fleet	IntentLog.commit: int-002	log	`0xd2587e57…cb4c`	22	38,104	success	hash matches, also on Base IntentLog
7	fleet	IntentLog.commit: int-003	log	`0x5cd73247…6b8f`	22	38,104	success	hash matches, also on Base IntentLog
8	fleet	IntentLog.commit: int-004	log	`0x639b790e…983a`	22	38,104	success	hash matches, also on Base IntentLog
9	fleet	anchor: charging 60 yuan for the flight	energy	`0xbebc5a07…2485`	22	38,104	success	
10	fleet	IntentLog.commit: int-005	log	`0x5946b78d…08ed`	23	38,104	success	hash matches, also on Base IntentLog
11	fleet	anchor: fee 1500 yuan released to the drone wallet	fee_in	`0x191e3afe…c9f0`	23	38,104	success	
12	fleet	anchor: rail fee 30 yuan	rail	`0x2746fd1b…e6bb`	23	38,104	success	
13	fleet	anchor: hold 199.95 yuan kept for 24 hours	hold	`0xc2ea3baa…4f7a`	23	38,104	success	
14	fleet	IntentLog.commit: int-006	log	`0xea6804b4…e694`	23	38,104	success	hash matches, also on Base IntentLog
15	fleet	anchor: instalment 600 yuan to the lender	instalment	`0x8fe29bd8…bc5e`	23	38,104	success	
16	fleet	anchor: pilot bond filled by 150 yuan from the fee	bond_fill	`0x2cabde12…169a`	23	38,104	success	
17	fleet	anchor: fee 250 yuan locked for Spray 50 mu of rice, field north	lock	`0x5a2b0f04…eb16`	24	38,104	success	
18	fleet	IntentLog.commit: int-007	log	`0x3859c23d…fd57`	24	38,104	success	hash matches, also on Base IntentLog
19	fleet	IntentLog.commit: int-008	log	`0xeff003be…b25e`	24	38,104	success	hash matches, also on Base IntentLog
20	fleet	IntentLog.commit: int-009	log	`0xc32cf36c…880e`	24	38,104	success	hash matches, also on Base IntentLog
21	fleet	anchor: charging 30 yuan for the flight	energy	`0xf33d3d60…f755`	24	38,104	success	
22	fleet	anchor: fee 250 yuan held after a flag	refund	`0xe6ce2eef…f385`	25	38,104	success	
23	fleet	IntentLog.commit: int-010	log	`0x633425f9…3af5`	26	38,104	success	hash matches, also on Base IntentLog
24	fleet	anchor: vendor bond slashed 200 yuan for PayeeNotAllowlisted, intent int-010	slash	`0xb72c68e7…6ada`	27	38,104	success	
25	fleet	anchor: fee 240 yuan locked for Spray 60 mu of rice, field north	lock	`0xb7906965…fce2`	28	38,104	success	
26	fleet	IntentLog.commit: int-011	log	`0xbda6759b…8b7e`	28	38,104	success	hash matches, also on Base IntentLog
27	fleet	IntentLog.commit: int-012	log	`0x4425035a…77c7`	28	38,104	success	hash matches, also on Base IntentLog
28	fleet	IntentLog.commit: int-013	log	`0xebd7367c…69bd`	28	38,104	success	hash matches, also on Base IntentLog
29	fleet	anchor: charging 36 yuan for the flight	energy	`0x18794ce6…0b1d`	28	38,104	success	
30	fleet	IntentLog.commit: int-014	log	`0x65f2ea15…2e40`	29	38,104	success	hash matches, also on Base IntentLog
31	fleet	anchor: fee 240 yuan released to the drone wallet	fee_in	`0x55b65ce3…c075`	29	38,104	success	
32	fleet	anchor: rail fee 4.8 yuan	rail	`0x633fa931…a0bb`	29	38,104	success	
33	fleet	anchor: hold 31.99 yuan kept for 24 hours	hold	`0x5f985f0a…26e8`	29	38,104	success	

#	Scenario	Event	Kind	Transaction	Block	Gas	Status	IntentLog check
34	fleet	IntentLog.commit: int-015	log	`0x9459ea49…3490`	29	38,104	success	hash matches, also on Base IntentLog
35	fleet	anchor: instalment 50 yuan to the lender	instalment	`0x8a021c6b…888a`	29	38,104	success	
36	fleet	anchor: pilot bond filled by 24 yuan from the fee	bond_fill	`0x2c81b006…3523`	29	38,104	success	
37	fleet	anchor: title transferred from the lender to the pilot	title	`0x2119e8ac…e8f3`	29	38,104	success	
38	fleet	anchor: hold 199.95 yuan released, no flag	release	`0xca6fb8d8…b2c2`	30	38,104	success	
39	fleet	anchor: hold 31.99 yuan released, no flag	release	`0x37d1d533…d484`	30	38,104	success	
40	neighbour	anchor: vendor bond of 5000 yuan posted	vendor_bond	`0x06613835…bb2e`	31	38,104	success	
41	neighbour	anchor: fee 60 yuan locked for Spray 20 mu of maize, field north	lock	`0xe15c24e2…3f8d`	32	38,104	success	
42	neighbour	IntentLog.commit: int-016	log	`0x10c7305e…6d33`	32	38,104	success	hash matches, not in Base capture
43	neighbour	anchor: job cancelled, fee 60 yuan back to the farmer	refund	`0x94ffca34…a2bb`	32	38,104	success	
44	neighbour	anchor: fee 1500 yuan locked for Spray 100 mu of passion fruit, field north	lock	`0x08b379f9…1c63`	33	38,104	success	
45	neighbour	IntentLog.commit: int-017	log	`0xb14c3023…37c9`	33	38,104	success	hash matches, not in Base capture
46	neighbour	IntentLog.commit: int-018	log	`0x0e5244cc…5ab7`	33	38,104	success	hash matches, not in Base capture
47	neighbour	IntentLog.commit: int-019	log	`0xde358349…a3e7`	33	38,104	success	hash matches, not in Base capture
48	neighbour	anchor: charging 60 yuan for the flight	energy	`0xb0556c70…d9a5`	33	38,104	success	
49	neighbour	IntentLog.commit: int-020	log	`0x926e309b…ddbb`	34	38,104	success	hash matches, not in Base capture
50	neighbour	anchor: fee 1500 yuan released to the drone wallet	fee_in	`0xe650fb4e…2807`	34	38,104	success	
51	neighbour	anchor: rail fee 30 yuan	rail	`0x08d4bbc5…6e67`	34	38,104	success	
52	neighbour	anchor: hold 199.95 yuan kept for 24 hours	hold	`0x58847d86…da4c`	34	38,104	success	
53	neighbour	IntentLog.commit: int-021	log	`0x340e788f…c7df`	34	38,104	success	hash matches, not in Base capture
54	neighbour	anchor: instalment 600 yuan to the lender	instalment	`0xe45d85cb…a29f`	34	38,104	success	
55	neighbour	anchor: pilot bond filled by 150 yuan from the fee	bond_fill	`0xc0c71024…75ce`	34	38,104	success	
56	neighbour	anchor: fee 250 yuan locked for Spray 50 mu of rice, field north	lock	`0x71e44d60…d321`	35	38,104	success	
57	neighbour	IntentLog.commit: int-022	log	`0xf956e4a6…537e`	35	38,104	success	hash matches, not in Base capture
58	neighbour	IntentLog.commit: int-023	log	`0x6cf8a83a…5543`	35	38,104	success	hash matches, not in Base capture
59	neighbour	IntentLog.commit: int-024	log	`0x5c804a79…142f`	35	38,104	success	hash matches, not in Base capture
60	neighbour	anchor: charging 30 yuan for the flight	energy	`0x51248f17…8a38`	35	38,104	success	
61	neighbour	anchor: fee 250 yuan held after a flag	refund	`0x098d1d2a…a087`	36	38,104	success	
62	neighbour	IntentLog.commit: int-025	log	`0x0b05ac49…8727`	37	38,104	success	hash matches, not in Base capture

#	Scenario	Event	Kind	Transaction	Block	Gas	Status	IntentLog check
63	neighbour	anchor: vendor bond slashed 200 yuan for PayeeNotAllowlisted, intent int-025	slash	0xe83562a7…da92	38	38,104	success	
64	neighbour	anchor: fee 240 yuan locked for Spray 60 mu of rice, field north	lock	0x83361246…695c	39	38,104	success	
65	neighbour	IntentLog.commit: int-026	log	0x6d6e3834…e97b	39	38,104	success	hash matches, not in Base capture
66	neighbour	IntentLog.commit: int-027	log	0x6873450e…e310	39	38,104	success	hash matches, not in Base capture
67	neighbour	IntentLog.commit: int-028	log	0xce7c64b2…fe49	40	38,104	success	hash matches, not in Base capture
68	neighbour	anchor: charging 36 yuan for the flight	energy	0xce89963c…d8c8	40	38,104	success	
69	neighbour	IntentLog.commit: int-029	log	0x507858a3…861a	41	38,104	success	hash matches, not in Base capture
70	neighbour	anchor: fee 240 yuan released to the drone wallet	fee_in	0xcd06b536…d8bd	41	38,104	success	
71	neighbour	anchor: rail fee 4.8 yuan	rail	0x7b95ff46…c12c	41	38,104	success	
72	neighbour	anchor: hold 31.99 yuan kept for 24 hours	hold	0xdd5c2126…2f98	41	38,104	success	
73	neighbour	IntentLog.commit: int-030	log	0x4cc70fbd…f280	41	38,104	success	hash matches, not in Base capture
74	neighbour	anchor: instalment 50 yuan to the lender	instalment	0x14a3f98e…9ad6	41	38,104	success	
75	neighbour	anchor: pilot bond filled by 24 yuan from the fee	bond_fill	0xe38e1e32…49dd	41	38,104	success	
76	neighbour	anchor: title transferred from the lender to the pilot	title	0x3d515ee1…4cb6	41	38,104	success	
77	neighbour	anchor: hold 199.95 yuan released, no flag	release	0xc4e7f10f…7e31	42	38,104	success	
78	neighbour	anchor: hold 31.99 yuan released, no flag	release	0xf6072ae1…9d6c	42	38,104	success	

4. FATE model provenance

The credit model trained with WeBank's FATE federated learning framework is pinned on the same consortium ledger: its sha256 (from `fate/model.json`) was committed to the FISCO IntentLog under the job id keccak256("fate-model"). Anyone holding the model file can recompute its canonical hash (every field except sha256 and sha256_of, sorted keys, no spaces; `pnpm test` does this) and check the ledger before trusting a credit score.

Field	Value
Model sha256	0x3f321ae68cdfed7e1eb4d01fc6b41b30dc2851d20e51d80df75c8a5343de314f
Job id	0xe83832c97db5735e2627f5e1f29531b7d08a724c2f00f81cad56609ab2dd6b6a (keccak256 of "fate-model")
Transaction	0x6a6be28d9178e3ab3403acb4e54e341cfb8119cc4e4c30aff1c0ba32d01da2e6
Block	82
Status	success
Read back now	committedAt 1792497335845, jobOf matches: yes, equals `web/src/engine/credit/fate-model.json`: yes

5. One real run on both ledgers

`web/evidence/run-2026-10-20T11-58-26-050Z.json`: the fleet scenario in real mode with the funded Base Sepolia key and the FISCO target both set. Each intent was sent to both chains in the same step. Its hash was already in the Base IntentLog from the earlier Base run (docs/EXECUTION_EVIDENCE.md), so on Base the step anchored it as calldata, as the engine does on any repeat, while on FISCO the same hash was committed to the IntentLog. 15 of 15 pairs have a successful receipt on both chains and the same intent hash on FISCO.

#	Intent	Intent hash	Base Sepolia	FISCO BCOS	Check
1	anchor: int-001	0x4f03d59d…7dcd	0x3e5f77c6…ea78 already in the Base log, anchored, success	0x0c4ca53c…6d1d block 83, success	same hash
2	anchor: int-002	0x17cd5f8e…ff25	0x7c22d674…7d3b already in the Base log, anchored, success	0xdb3a75c5…a6d3 block 84, success	same hash

#	Intent	Intent hash	Base Sepolia	FISCO BCOS	Check
3	anchor: int-003	0x23127228…701e	0x4fde8e89…111a already in the Base log, anchored, success	0x20b87bca…ce36 block 85, success	same hash
4	anchor: int-004	0xd04a71ea…593f	0xa0b0fa2d…64f7 already in the Base log, anchored, success	0xbcb1e2d8…0515 block 86, success	same hash
5	anchor: int-005	0xc6918cd1…c75e	0xec6728f6…c0eb already in the Base log, anchored, success	0x2ae62b0c…a908 block 87, success	same hash
6	anchor: int-006	0x6ddcbed2…2ed9	0xc0475255…7284 already in the Base log, anchored, success	0xd84cfa34…2cd2 block 88, success	same hash
7	anchor: int-007	0x6ffe902d…f642	0xfa2dc7f4…aba8 already in the Base log, anchored, success	0xf9041a33…dbcb block 89, success	same hash
8	anchor: int-008	0xfc0d28bd…1f58	0x4733c619…1a9d already in the Base log, anchored, success	0x9182aa3e…f694 block 90, success	same hash
9	anchor: int-009	0x84889a9e…041f	0x955b746c…e05c already in the Base log, anchored, success	0x9893a8e3…db32 block 91, success	same hash
10	anchor: int-010	0xc175b4f4…4432	0x9b41e5f2…0158 already in the Base log, anchored, success	0xfc01cb30…caf1 block 92, success	same hash
11	anchor: int-011	0x87220bc9…4127	0x08d96ffc…455a already in the Base log, anchored, success	0xccb5852e…4b39 block 93, success	same hash
12	anchor: int-012	0x5d6bcc1e…d28c	0x8b0fd4ab…47db already in the Base log, anchored, success	0x549ae385…e809 block 94, success	same hash
13	anchor: int-013	0xf52c5b2f…efc6	0x607a6e24…13ad already in the Base log, anchored, success	0xf4a28815…5e72 block 95, success	same hash
14	anchor: int-014	0x58f0d784…eefa	0xd880e75d…d048 already in the Base log, anchored, success	0xbc6d89da…d1fd block 96, success	same hash
15	anchor: int-015	0xbbaeb1d5…937a	0xe885c19f…59ca already in the Base log, anchored, success	0x3ef35d9a…a4e8 block 97, success	same hash

6. One full job against the deployed contracts

Run by `contracts/script/fulljob.sh fisco`: fresh keys for six roles, every step a call to the deployed Bond, Lease, JobBoard or IntentLog with native value moving: vendor bond posted, fee locked, confirmed and split (instalment to the lender through Lease), a rogue intent committed and slashed by proof from the vendor bond, the last payment moving the title to the pilot, the hold released. 10 of 10 receipts fetched again now with status success.

#	Call	From	Value	Transaction	Block	Status
1	Bond.postVendorBond	vendor	20000000000 wei	0x9387f7e88f57d80543b055506cf0ca2cd3cea937b1907fe746240a0ca8cdb1e4	104	success
2	Bond.register	agent	0	0x562e77037b96208e4658f71be3c8b47bb8794c75311419930699ae80c20a7bd0	105	success
3	Lease.open	lender	0	0xaa04273623fee61b841c24d8881edf263d3c6019636e871a11603bb653a50360	106	success
4	JobBoard.post	farmer	100000000000 wei	0xd37022a1a1855e618bf43296b30758f665a0d53e11767576cb5265e9ebd2963b	107	success
5	JobBoard.start	agent	0	0x5233a4ddfca6ccea96ff66ddec2d4ac5731932c4ec4b35b565144f3cdc8f1be6	108	success
6	JobBoard.confirm	verifier	0	0x666671bf2a494e4989c98bd4048c7335aba378d116ecdd06c1c3fc3f73e174d5	109	success
7	IntentLog.commit	agent	0	0xa407d8c072f8565c31e139039636c5a586e2dffcbf7adf463e3003b53bfc130e	110	success
8	Bond.slashByProof	lender	0	0x3cfcbadc0159b7b612683fc870a2e1e231674d08a67abdfc5fbaf638cc2c53df	111	success
9	Lease.pay	pilot	60000000000 wei	0x48aaf9d2866c3605b2c117bbab38ae42171bab9897b2d969fca785e349238665	112	success
10	JobBoard.releaseHold	agent	0	0xaed2ba7e0e4f2cc51ef162b2dd9ead9c790e128a4f9cbbfc91876d9cac0f3c3d	113	success

7. Reproduce

```
bash contracts/script/fisco.sh     # node in Docker, deploy, smoke, FATE model hash, contracts/deployments/fisco-bcos.json
cd web
export FISCO_RPC_URL=http://127.0.0.1:8545 FISCO_PRIVATE_KEY=$(cat ../contracts/.fisco/deployer.key)
export FISCO_INTENT_LOG=$(node -p "require('../contracts/deployments/fisco-bcos.json').contracts.IntentLog.address")
SURETY_PRIVATE_KEY= JIXIN_REAL=1 pnpm evidence   # FISCO only, lands in evidence/fisco/
pnpm fisco                         # receipts, IntentLog read back, this file
```